

Week 3 Deliverable: Preliminary Lunar Regolith Transport Rover Concepts

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1 Week 3 Objective

The objective of Week 3 was to transition from general lunar rover research into preliminary rover architecture and mission planning. The main focus was developing rough hand-drawn concept sketches, identifying the major rover subsystems, defining the mission sequence, and beginning to evaluate how a 6-wheel rocker-bogie lunar regolith transport rover could support a 750 lb payload.

At this stage, the design is still conceptual. Final dimensions, material selections, motor sizing, and detailed structural analysis have not yet been completed. The goal of this deliverable is to establish the basic system layout and design direction before moving into detailed CAD modeling and engineering calculations.

2 Mission Statement

The proposed rover is a semi-autonomous lunar regolith transport rover designed to move loose regolith from a collection zone to a lunar habitat shielding zone. The rover is intended to support the construction of radiation shielding by transporting regolith to trays, berms, or shielding structures around a lunar surface habitat.

The rover is designed for a target payload of approximately:

$$750 \text{ lb} \approx 340 \text{ kg}$$

The rover is not currently intended to perform full excavation by itself. Instead, regolith may be loaded by astronauts, a separate excavation rover, or a bucket/scoop loading system that can be developed separately. This allows the rover concept to focus primarily on transportation, mobility, unloading, and system integration.

Based on the Week 2 payload trade study, a 340 kg payload was selected as an intermediate design target between the 250 kg baseline rover and the 450 kg heavy-hauler case. This value reduces total trip count compared to the 250 kg case while avoiding the added structural and stability challenges of the full 450 kg heavy-hauler concept.

3 Mission Requirements

Table 1: Preliminary mission requirements for the lunar regolith transport rover.

Requirement	Preliminary Target
Primary mission	Transport lunar regolith for habitat shielding
Payload capacity	750 lb, approximately 340 kg
Mobility system	6-wheel rocker-bogie suspension
Operation type	Semi-autonomous or remotely supervised
Loading method	Astronaut loading, separate excavation rover, bucket, or scoop system
Unloading method	Rear-hinged dump bed with actuator-assisted release
Terrain	Loose regolith, rocks, slopes, and uneven lunar surface
Power system	Battery system with deployable solar panels
Navigation	Cameras, sensors, onboard computer, and communication link
Environmental protection	Dust protection, sealed electronics, and protected joints

4 Mission Operation Sequence

The basic mission sequence is shown below:

1. Leave charging station.
2. Navigate to regolith collection zone.
3. Load regolith into rover bed.
4. Drive to habitat shielding zone.
5. Dump regolith into trays or berms.
6. Return to collection zone or charging station.
7. Repeat until shielding requirement is complete.

5 Mission Operation Flowchart

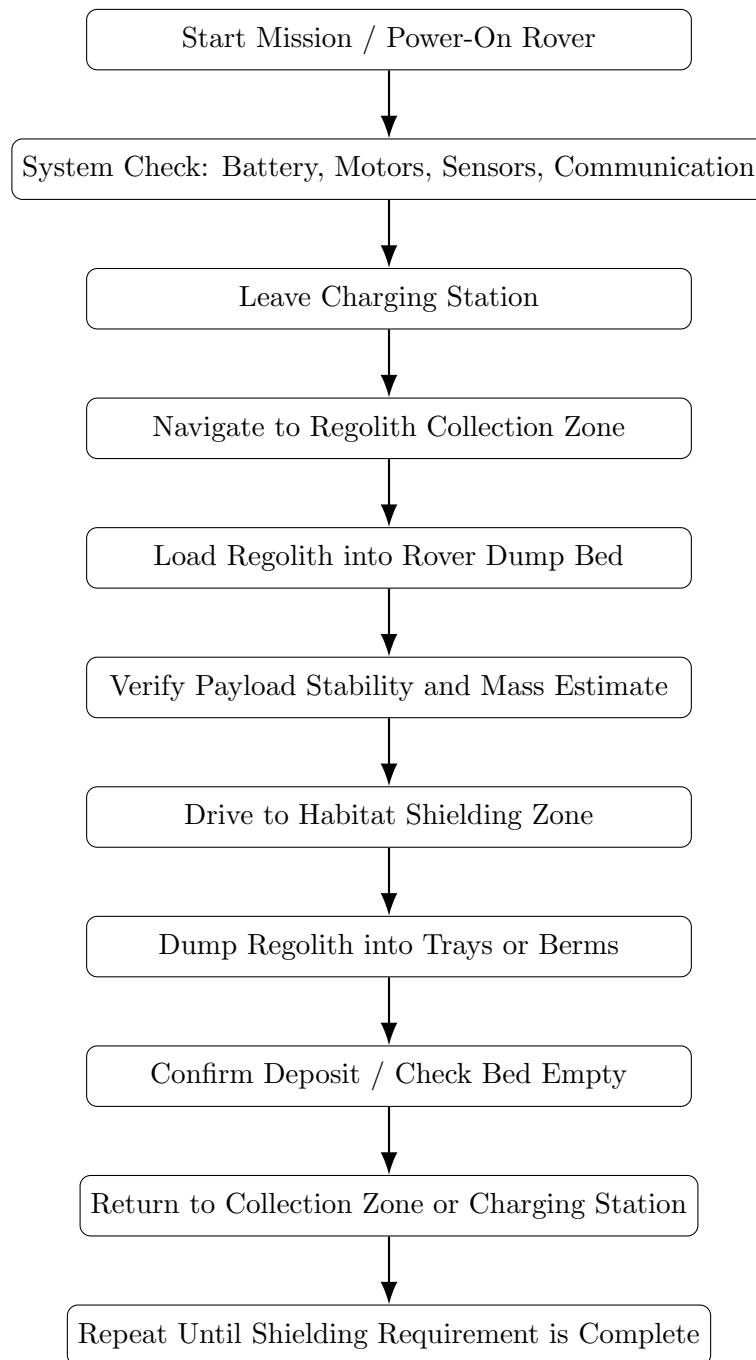


Figure 1: Preliminary mission operation flowchart for the lunar regolith transport rover.

6 Basic Lunar Base Route Layout

The rover mission can be simplified into three main zones: the charging station, the regolith collection zone, and the habitat shielding zone. The rover repeatedly travels between these

areas to support construction of regolith-based radiation shielding.

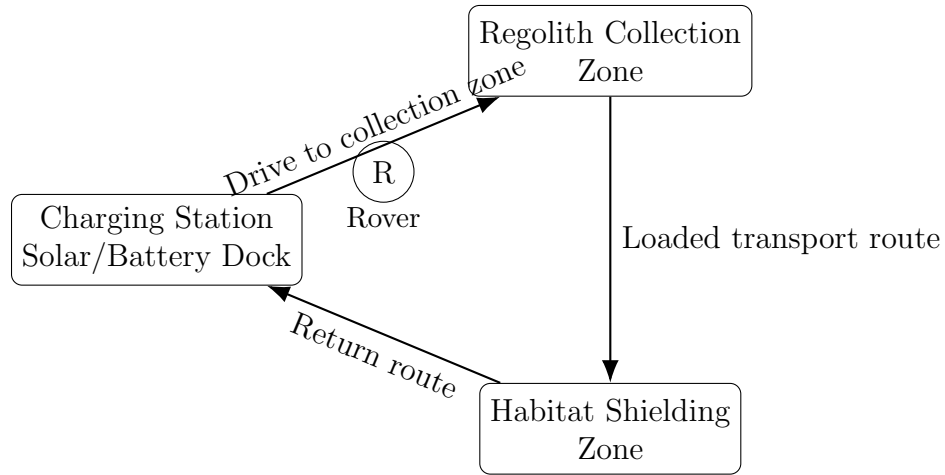


Figure 2: Basic lunar base route layout showing rover movement between charging, collection, and shielding zones.

7 Subsystem Breakdown

The rover is divided into multiple major subsystems. Each subsystem must be developed separately, but all must work together to complete the regolith transport mission.

Table 2: Preliminary subsystem breakdown.

Subsystem	Function and Major Components
Chassis / Frame	Main structural body that supports the dump bed, battery, electronics, suspension mounts, and payload loads. Should include cross-members and reinforced mounting points.
Wheels and Mobility System	6-wheel rocker-bogie system with three wheels per side. Includes wheel hubs, wheel motors, rocker arms, bogie links, pivot joints, and wheel brackets.
Dump Bed	Holds regolith during transport. Uses a rear-hinged bed and actuator-assisted lifting mechanism to unload material into trays or berms.
Bucket, Scoop, or Loading Method	Defines how regolith enters the rover bed. Current concept allows astronaut loading, a separate excavation rover, or a future bucket/scoop attachment.
Battery System	Provides electrical power to the rover. Battery should be mounted low in the chassis to reduce the center of gravity and improve stability.
Sensors	Cameras, IMU, wheel encoders, LiDAR/radar, hazard detection sensors, and possible load sensors for payload estimation.
Onboard Computer	Controls navigation, motor commands, actuator control, sensor processing, communication, and mission logic.
Communication System	Antenna, transmitter/receiver, and communication link to habitat, lander, astronaut operator, or relay system.
Dust Protection	Sealed electronics box, protected cable routing, dust covers over joints, shielded actuator rod, and protected motor housings.

8 Subsystem Breakdown Diagram

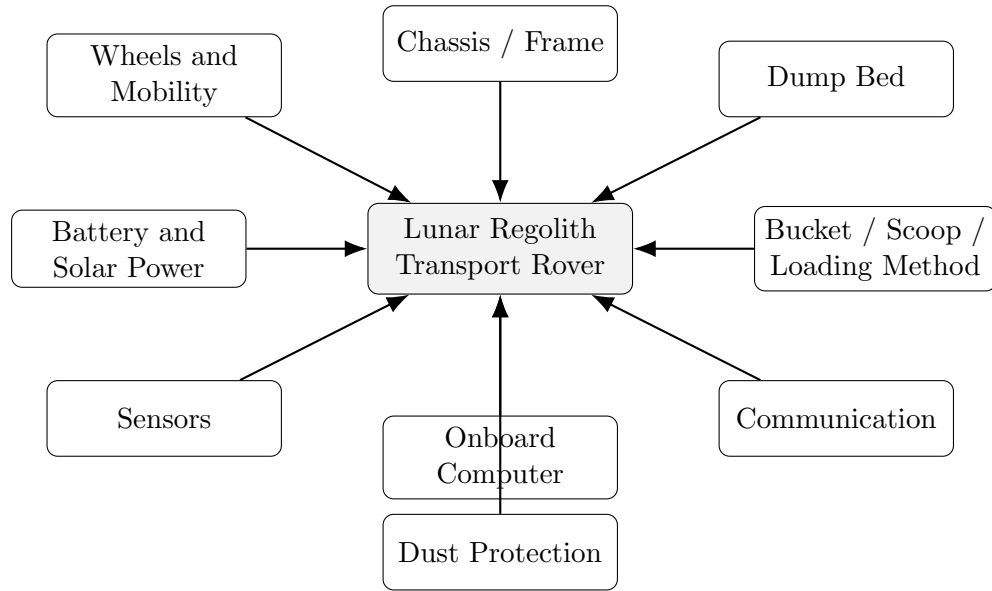


Figure 3: Subsystem breakdown diagram for the rover concept.

9 Preliminary Rover Concept

The selected early design direction is a 6-wheel rocker-bogie lunar regolith transport rover with a rear-hinged dump bed. The rover is designed around a 750 lb payload and includes a low-mounted battery system, deployable solar panels, a sensor/communication mast, and a protected electronics enclosure.

The rocker-bogie suspension is used because it allows the rover to travel over uneven lunar terrain while distributing loads across multiple wheels. In side view, only three wheels are visible, but the full rover contains six total wheels, with three wheels on each side.

9.1 Rocker-Bogie Load Path

The intended load path is:

Regolith Payload → Dump Bed → Main Chassis Frame → Rocker Pivot Brackets → Rocker-Bogie Arms →

This load path is important because the dump bed should be mounted to the main chassis frame rather than directly to the moving suspension arms. The suspension arms must remain free to rotate over terrain.

9.2 Rocker-Bogie Concept Sketch

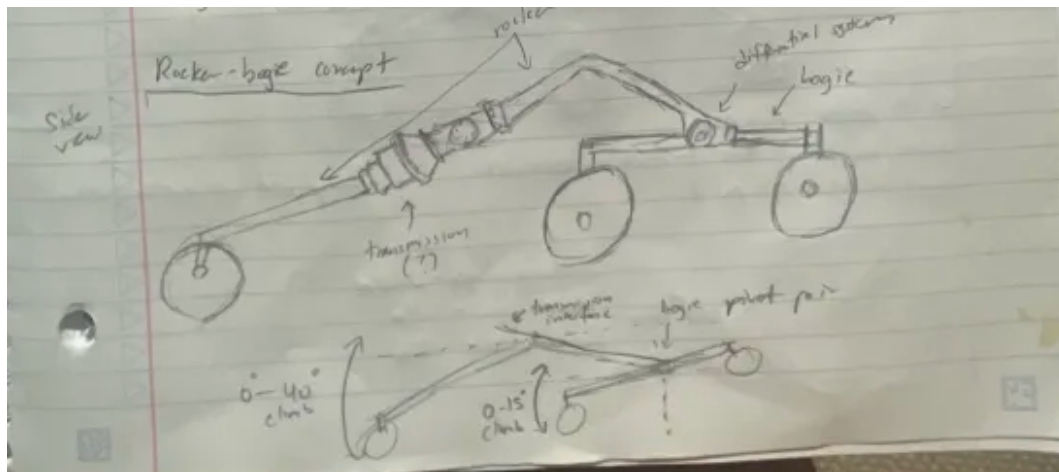


Figure 4: Full preliminary rover concept sketch.

9.3 Wheel Concept Sketch

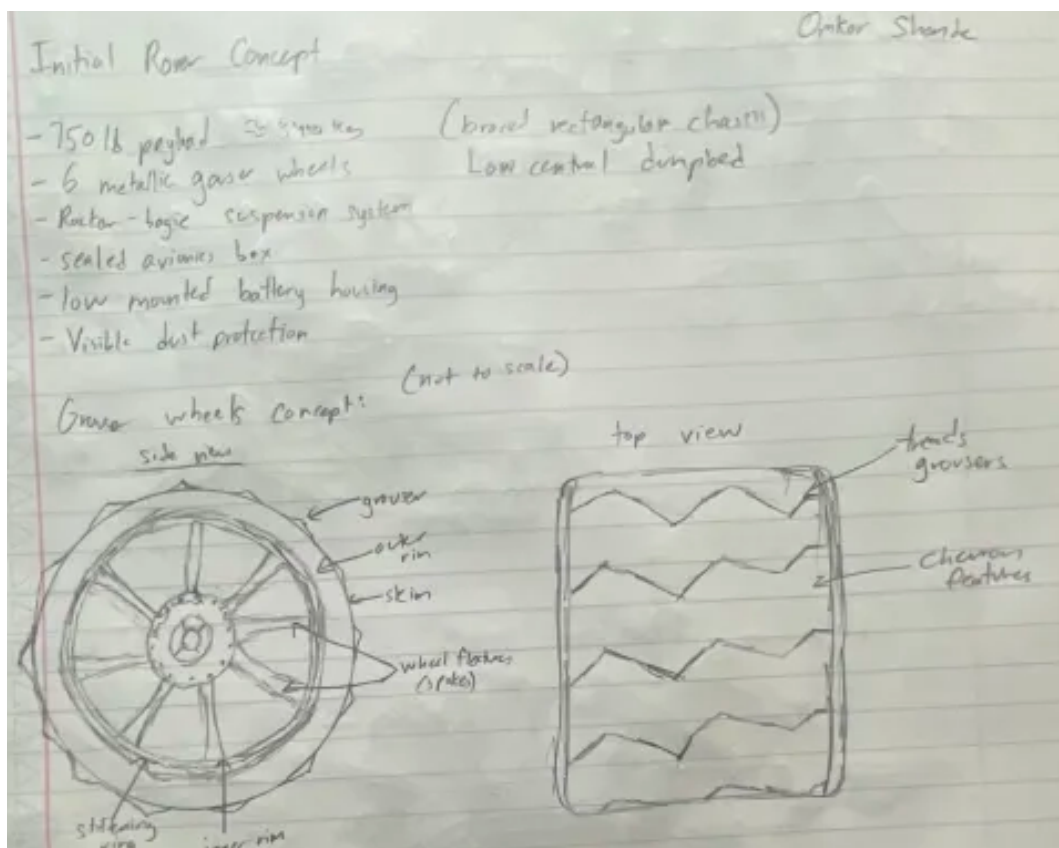


Figure 5: Full preliminary rover concept sketch.

9.4 Full Rover Concept Sketch

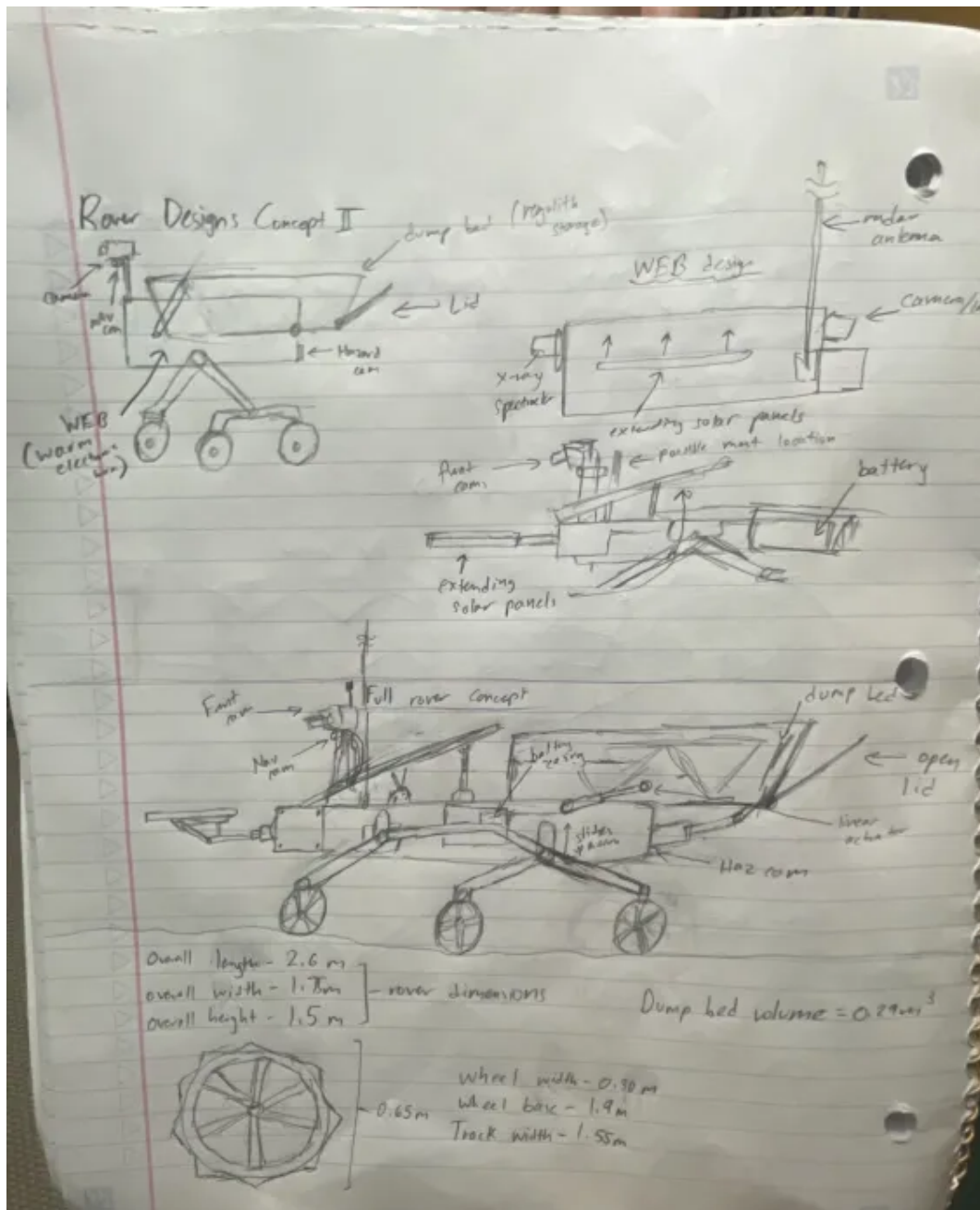


Figure 6: Full preliminary rover concept sketch.

9.5 Dump Bed Mechanism Sketch

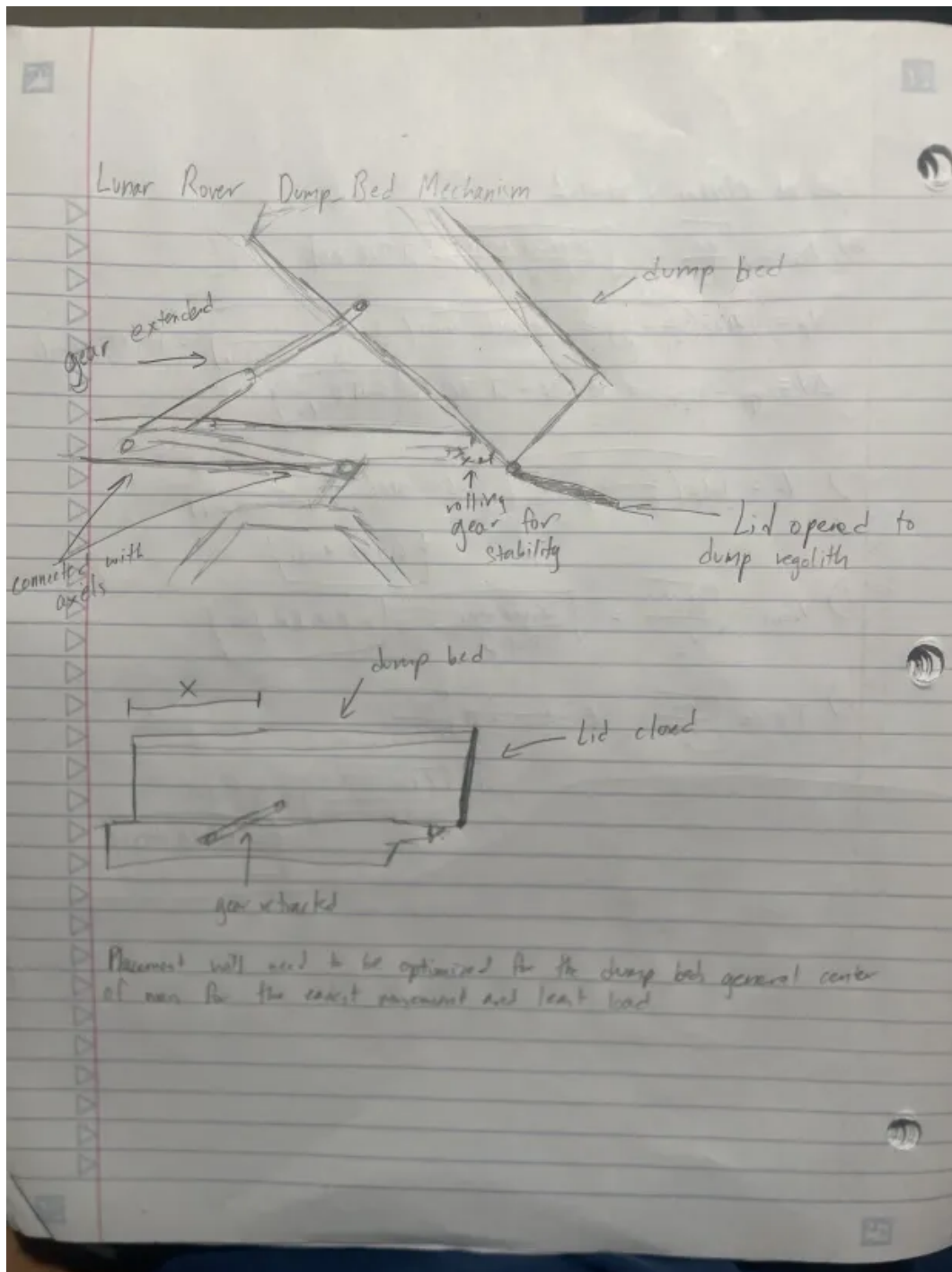


Figure 7: Full preliminary rover concept sketch.

10 Dump Bed Mechanism

The dump bed is intended to unload regolith after the rover reaches the habitat shielding zone. The bed remains flat during transport and is lifted during unloading using a linear actuator. The dump bed rotates about a rear hinge, causing regolith to slide out through a passive or controlled tailgate.

The proposed dumping angle is approximately:

$$45^{\circ} - 50^{\circ}$$

This angle is selected as an early estimate because loose granular material generally requires a steep enough slope to slide out reliably. The dumping operation should only occur while the rover is stationary and on relatively flat ground to reduce tipping risk.

11 Preliminary Design Dimensions

The following dimensions are early estimates and are not final. They are included to guide future CAD modeling and load calculations.

Table 3: Preliminary rover dimensions.

Parameter	Preliminary Estimate
Payload capacity	750 lb, approximately 340 kg
Overall length	2.5–3.0 m
Overall width	1.5–2.0 m
Overall height	1.3–1.7 m
Wheel diameter	0.45–0.65 m
Number of wheels	6 total, 3 per side
Dump bed angle	45–50 degrees
Operation mode	Semi-autonomous / supervised

12 Engineering Concerns Identified

Several engineering concerns were identified during the Week 3 concept phase:

- The rocker-bogie members cannot be too long or too thin because long members increase bending moment.
- Suspension arms should use rectangular hollow beams or reinforced box-section members instead of thin rods.
- Pivot joints should use reinforced brackets, gussets, and double-shear mounting where possible.
- The battery should be mounted low in the chassis to lower the center of gravity.

- The dump bed should be mounted to the chassis, not directly to the suspension arms.
- The rover should have a wide enough track width to remain stable while carrying and dumping regolith.
- Dust protection is critical for joints, wheel motors, actuator rods, electronics, and cable routing.
- Deployable solar panels should be folded or protected during rough driving and deployed mainly during charging or low-speed operations.